

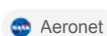
Yes, your understanding is partly right: **count/m³ from the particle counter is particle number per sampled air volume**. But the AERONET plot is **not number concentration**. It is **aerosol particle volume distribution**:

$$\frac{dV(r)}{d \ln r}$$

with units:

$$\mu m^3 / \mu m^2$$

That means **particle volume per atmospheric column area**, not per cubic meter of local air. AERONET retrieves this for radius r , not diameter, over roughly $0.05 \leq r \leq 15 \mu m$.



AERONET also states that the retrieved volume distribution relates to number distribution by

$$\frac{dV(r)}{d \ln r} = \frac{4\pi}{3} r^3 \frac{dN(r)}{d \ln r}$$

and that the volume concentration is for the **total atmospheric column**.



1. First: make sure your counter data are differential, not cumulative

Your GT-526S particle counter can output total counts, particles/L, particles/ft³, or particles/m³. It also has **Normal** mode and **Difference** mode. In Normal mode, the counts are cumulative: for example, the 0.5 μm channel means particles **0.5 μm and larger**. In Difference mode, each channel is a bin between one size and the next larger size.



If your data are cumulative:

$$C_{\geq D_i}$$

then convert to differential bins:

$$\Delta N_i = C_{\geq D_i} - C_{\geq D_{i+1}}$$

Example for diameter bins:

$$N_{0.3-0.5} = C_{\geq 0.3} - C_{\geq 0.5}$$

$$N_{0.5-1.0} = C_{\geq 0.5} - C_{\geq 1.0}$$

and so on.

If the instrument is already in **Difference** mode, then you can skip this subtraction.

2. Convert diameter to radius

The particle counter channels are usually in **diameter**:

$$D = 0.3, 0.5, 1.0, 2.0, 5.0, 10.0 \mu m$$

AERONET uses **radius**, so:

$$r = \frac{D}{2}$$

Your radius bin edges become:

$$0.15, 0.25, 0.50, 1.00, 2.50, 5.00 \mu m$$

Use the geometric bin center:

$$r_{mid,i} = \sqrt{r_i r_{i+1}}$$

and the logarithmic bin width:

$$\Delta \ln r_i = \ln \left(\frac{r_{i+1}}{r_i} \right)$$

Since $D = 2r$, you can also use:

$$\Delta \ln r = \Delta \ln D$$

because the factor of 2 cancels.

3. Convert count/m³ to local volume distribution

For each bin:

$$\frac{dN}{d \ln r} \approx \frac{\Delta N_i}{\Delta \ln r_i}$$

where ΔN_i has units of count/m³.

Then convert number distribution to volume distribution:

$$\left(\frac{dV}{d \ln r} \right)_{local} = \frac{4\pi}{3} r_{mid,i}^3 \frac{\Delta N_i}{\Delta \ln r_i}$$

If r is in μm , then the local result has units:

$$\mu m^3 / m^3$$

This is the particle volume per cubic meter of sampled air. This is **not yet AERONET's unit**.

4. Convert local balloon data to AERONET-like column units

AERONET is column-integrated, so you need to integrate over altitude:

$$\left(\frac{dV}{d \ln r}\right)_{column} = \int \left(\frac{dV}{d \ln r}\right)_{local} dz$$

For balloon data, use a sum:

$$\left(\frac{dV}{d \ln r}\right)_{column} \approx \sum_j \frac{4\pi}{3} r_{mid,i}^3 \frac{\Delta N_{i,j}}{\Delta \ln r_i} \Delta z_j$$

This gives:

$$\mu m^3 / m^2$$

To match AERONET's $\mu m^3 / \mu m^2$, convert m^2 to μm^2 :

$$1 m^2 = 10^{12} \mu m^2$$

so:

$$\left(\frac{dV}{d \ln r}\right)_{AERONET-like} = 10^{-12} \sum_j \frac{4\pi}{3} r_{mid,i}^3 \frac{\Delta N_{i,j}}{\Delta \ln r_i} \Delta z_j$$

Final units:

$$\mu m^3 / \mu m^2$$

This is the best derivation for comparing your balloon particle counter result with the AERONET image.

Python-style calculation

<> Python



▶ Run

```
import numpy as np

# Particle counter diameter thresholds in micrometers
D_edges = np.array([0.3, 0.5, 1.0, 2.0, 5.0, 10.0])

# Convert to radius edges
r_edges = D_edges / 2

# Geometric radius centers
```

```

r_mid = np.sqrt(r_edges[:-1] * r_edges[1:])

# Log bin widths
dlnr = np.log(r_edges[1:] / r_edges[:-1])

# Example: cumulative concentrations in count/m^3
# Shape: altitude/time rows x size channels
# Ccum[:, 0] = count/m^3 for >=0.3 um
# Ccum[:, 1] = count/m^3 for >=0.5 um, etc.

# Convert cumulative to differential bins
dN = Ccum[:, :-1] - Ccum[:, 1:] # count/m^3

# Local volume distribution: um^3/m^3
dVdlnr_local = (4*np.pi/3) * r_mid**3 * (dN / dlnr)

# z_m is altitude in meters
# Integrate over altitude and convert m^2 to um^2
dVdlnr_column = 1e-12 * np.trapz(dVdlnr_local, z_m, axis=0)

```

Important comparison warning

If you only use **one balloon altitude**, you should compare only the **shape**, not the absolute y-values. For the absolute AERONET y-axis, you need the vertical integral through the atmospheric column.

Also, your counter size range is roughly:

$$D = 0.3 \text{ to } 10 \mu m$$

or

$$r = 0.15 \text{ to } 5 \mu m$$

but AERONET covers about:

$$r = 0.05 \text{ to } 15 \mu m$$

So your counter misses the smallest fine particles and some large coarse particles. That means the peaks may not match perfectly, even if your derivation is correct.